Welcome

Thank you for purchasing the MEAP for *Digital Twins in Action*.

Digital twins have been a hot topic for several years now and there is a lot of excitement, interest, and opportunity in the field. A lot of this is because the technology that enables you to build a digital twin has become accessible to just about anyone now, opening up endless possibilities for innovation.

Despite all the buzz, there's surprisingly little practical guidance available. Google "digital twins" and you'll find countless glossy brochures, white papers, and marketing materials filled with futuristic 3D renderings but often lacking real substance. When you try to go deeper, you quickly discover there's no comprehensive, practical guide explaining what a digital twin actually is and, more importantly, how you can build one yourself. This gap isn't surprising. Unlike well-established technologies like the Java programming language or relational databases which have dozens of excellent books dedicated to them, a digital twin isn't a single technology but rather a system integrating multiple advanced disciplines: the Internet of Things, computer graphics, databases, artificial intelligence, and machine learning, each complex enough to warrant its own book.

I'll admit it's a bit of a cliche, but I've genuinely tried to create the book I wish I'd had when starting my digital twin journey. You'll explore all the essential topics for building a digital representation of a physical system—from creating a simple sensor to measure real-world changes, all the way through programming an AI agent to make autonomous decisions. I'll provide practical examples throughout and take you along as I build a digital twin of my own home.

While you'll need some familiarity with Python to follow the practical examples, and TypeScript for the frontend components, I believe you'll gain valuable insights even if you choose not to follow every line of code! I